

EE2703 - Applied Programming Lab (Report 2)

Nithin G - EE24B046

1. Assignment Overview

The assignment had the following five functions to implement and we will go through each of those.

2. `find_temperature_extremes()`

This function aims to find the hottest and coldest months for a given city.

- The function finds the temperature for every month of a city.
- Then using

```
max(temperature_data, key=temperature_data.get) ;  
min(temperature_data, key=temperature_data.get)
```

we will be able to get the hottest and coldest temperature months of the city.

3. `get_seasonal_averages()`

This function aims to calculate the seasonal temperature averages of a city for all the years wherever the data is not **None**

4. `compare_decades()`

This function aims to compare the average between 2 decades for a function and comments on its trend between the decades.

If the

```
average_temperature_1 < average_temperature_2
```

then the trend is **warming**. Else if the inequality is reversed it returns **cooling** and if equality holds it returns **stable**

5. find_similar_cities()

This function first finds the average temperature for all the cities and then compares its average values with the **target_city** and then returns the cities which are having a **avg_temp_diff** lesser than the tolerance.

6. get_temperature_trends()

The yearly average was computed by **temp_yearly_average** and the moving average was computed by **temp_moving_average**

The overall slope was computed by

```
overall_slope = round(((q[-1][1] - q[0][1])/(len(q)-1)),3)
```

which is taken from the standard source <https://www.quantifiedstrategies.com/moving-average-slope>

Later the warming and cooling periods are computed by the **detect_warming_cooling_slopes** which returns the slope of warming and cooling periods.

7. Date vs Temperatures(get_city_temperatures())

